A Suboptimality Approach to Distributed H_2 Optimal Control by Dynamic Output Feedback

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Applications

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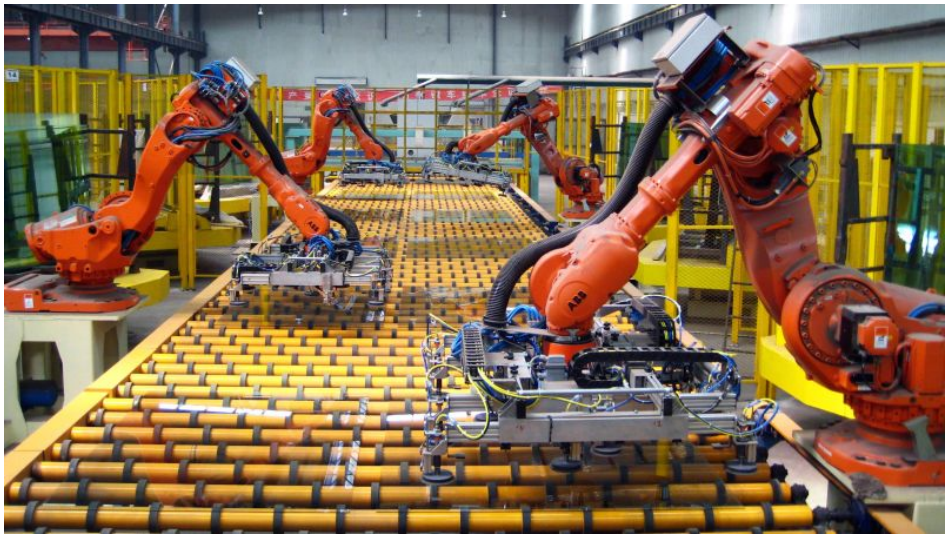

Figure: Synchronization of industrial robot arms

Motivation

In these applications, a truck or a robot, can be regarded as an agent. Together, these truck or robots form a multi-agent system.

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Motivated by the above, in this talk, we consider the problem of distributed H_2 optimal control for multi-agent systems.

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Distributed H_2 optimal control problem

Multi-agent system:

$$\begin{aligned} \dot{x}_i &= Ax_i + Bu_i + Ed_i, \\ y_i &= C_1 x_i + D_1 d_i, \qquad\qquad i = 1, 2, \ldots, N \\ z_i &= C_2 x_i + D_2 u_i, \end{aligned} \tag{1}$$

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Distributed protocol:

$$\begin{aligned} \dot{w}_i &= Aw_i + B\sum_{j=1}^{N} a_{ij}(u_i - u_j) + G\left(\sum_{j=1}^{N} a_{ij}(y_i - y_j) - C_1 w_i\right), \\ u_i &= Fw_i, \quad i = 1, 2, \ldots, N \end{aligned} \tag{2}$$

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Communication: connected weighted undirected graphs

Distributed H_2 optimal control problem

Denote

$$\mathbf{x} = (x_1^\top, x_2^\top, \ldots, x_N^\top)^\top, \qquad \mathbf{u} = (u_1^\top, u_2^\top, \ldots, u_N^\top)^\top,$$
$$\mathbf{y} = (y_1^\top, y_2^\top, \ldots, y_N^\top)^\top, \qquad \mathbf{z} = (z_1^\top, z_2^\top, \ldots, z_N^\top)^\top,$$
$$\mathbf{d} = (d_1^\top, d_2^\top, \ldots, d_N^\top)^\top, \qquad \mathbf{w} = (w_1^\top, w_2^\top, \ldots, w_N^\top)^\top.$$

Distributed H_2 optimal control problem

Multi-agent system:

$$\begin{aligned}\dot{\mathbf{x}} &= (I_N \otimes A)\mathbf{x} + (I_N \otimes B)\mathbf{u} + (I_N \otimes E)\mathbf{d},\\ \mathbf{y} &= (I_N \otimes C_1)\mathbf{x} + (I_N \otimes D_1)\mathbf{d},\\ \mathbf{z} &= (I_N \otimes C_2)\mathbf{x} + (I_N \otimes D_2)\mathbf{u}\end{aligned} \tag{3}$$

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Distributed protocol:

$$\begin{aligned}\dot{\mathbf{w}} &= (I_N \otimes (A - GC_1) + L \otimes BF)\,\mathbf{w} + (L \otimes G)\mathbf{y},\\ \mathbf{u} &= (I_N \otimes F)\mathbf{w}\end{aligned} \tag{4}$$

The matrix L is the Laplacian matrix of the given communication graph.

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Controlled network:

$$\begin{pmatrix} \dot{\mathbf{x}} \\ \dot{\mathbf{w}} \end{pmatrix} = \begin{pmatrix} I_N \otimes A & I_N \otimes BF \\ L \otimes GC_1 & I_N \otimes (A - GC_1) + L \otimes BF \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{w} \end{pmatrix} + \begin{pmatrix} I_N \otimes E \\ L \otimes GD_1 \end{pmatrix} \mathbf{d},$$

$$\mathbf{z} = \begin{pmatrix} I_N \otimes C_2 & I_N \otimes D_2 F \end{pmatrix} \begin{pmatrix} \mathbf{x} \\ \mathbf{w} \end{pmatrix}$$

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Foremost, we want the protocol (4) to achieve synchronization.

Definition 1

The protocol (4) is said to achieve synchronization if, whenever $\mathbf{d} = 0$, we have $x_i(t) - x_j(t) \to 0$ and $w_i(t) - w_j(t) \to 0$ as $t \to \infty$, for all $i, j = 1, 2, \ldots, N$.

Distributed H_2 optimal control problem

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These differences are captured by the incidence matrix R of the communication graph.

Distributed H_2 optimal control problem

We introduce a new output

$$\zeta = (W^{\frac{1}{2}} R^{\top} \otimes I_p)\mathbf{z},$$

where W is the weight matrix of the communication graph.

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Note that

$$L = RWR^\top.$$

Distributed H_2 optimal control problem

Controlled network with the new output:

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Impulse response:

$$T_{F,G}(t) = C_e e^{A_e t} E_e$$

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Impulse response:

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Cost functional:

$$J(F, G) := \int_0^\infty \operatorname{tr}\left[T_{F,G}{}^\top(t) T_{F,G}(t) \right] dt$$

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Distributed protocol:

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Distributed H_2 optimal control problem

Design local gains F and G such that the dynamic protocol (4) minimizes $J(F, G)$ while synchronizing the controlled network.

Distributed H_2 optimal control problem

Unfortunately,

due to the fact that the set of protocols over which the cost functional needs to be minimized is a non-convex set: the protocol (4) has a pre-specified structure, imposed by the network graph.

This optimization problem is, in general, non-convex and difficult to solve.

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Therefore, in this talk we will solve a version of this problem that only requires suboptimality.

Distributed H_2 suboptimal control problem

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Cost functional: $J(F, G) := \int_0^\infty \operatorname{tr}\left[T_{F,G}{}^\top(t)\, T_{F,G}(t)\right] dt$

Distributed H_2 suboptimal control problem

Let $\gamma > 0$. Design local gains F and G such that the protocol (4) achieves $J(F, G) < \gamma$ while synchronizing the network.

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Brief ideas

1. We first transform the distributed $\mathcal{H}_2$ suboptimal control problem for linear multi-agent systems by dynamic output feedback into the $\mathcal{H}_2$ suboptimal control problems for $N-1$ linear systems by dynamic output feedback with the same control gains F and G.

Brief ideas

1. We first transform the distributed $\mathcal{H}_2$ suboptimal control problem for linear multi-agent systems by dynamic output feedback into the $\mathcal{H}_2$ suboptimal control problems for $N-1$ linear systems by dynamic output feedback with the same control gains F and G.
2. As necessary preliminary results, we solve the open problem of finding, for continuous-time linear systems, a separation principle based H_2 suboptimal dynamic output feedback cotroller.

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2. As necessary preliminary results, we solve the open problem of finding, for continuous-time linear systems, a separation principle based H_2 suboptimal dynamic output feedback cotroller.
3. By using of the results in Step 2, as the main result, we provide a design method for computing local gains F and G such that the resulting distributed protocol guarantees $J(F,G) < \gamma$ and achieves synchronization.

Main result

Theorem 1

Let $\gamma > 0$. Assume that $D_1 E^\top = 0$, $D_2^\top C_2 = 0$, $D_1 D_1^\top = I_r$ and $D_2^\top D_2 = I_m$. Let $Q > 0$ satisfy

$$AQ + QA^\top - QC_1^\top C_1 Q + EE^\top < 0.$$

Let c be such that $0 < c < \frac{2}{\lambda_N^2}$. We distinguish two cases:

1. if $\frac{2}{\lambda_2^2+\lambda_2\lambda_N+\lambda_N^2} \leq c < \frac{2}{\lambda_N^2}$, then there exists $P > 0$ satisfying

$$A^\top P + PA + (c^2\lambda_N^3 - 2c\lambda_N)PBB^\top P + \lambda_N C_2^\top C_2 < 0.$$

2. if $0 < c < \frac{2}{\lambda_2^2+\lambda_2\lambda_N+\lambda_N^2}$, then there exists $P > 0$ satisfying

$$A^\top P + PA + (c^2\lambda_2^3 - 2c\lambda_2)PBB^\top P + \lambda_N C_2^\top C_2 < 0.$$

The four assumptions are made to simplify the notation, and can be replaced by $D_1 D_1^\top < 0$ and $D_2^\top D_2 > 0$ alone.

Theorem 1 (continued)

In both cases, if in addition P and Q satisfy

$$\operatorname{tr}\left(C_1 QPQ C_1^\top\right) + \lambda_N \operatorname{tr}\left(C_2 Q C_2^\top\right) < \frac{\gamma}{N-1}.$$

Then the protocol (2) with $F := -cB^\top P$ and $G := QC_1^\top$ synchronizes the network and it is suboptimal, i.e. $J(F, G) < \gamma$.

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- We have formulated the distributed H_2 suboptimal control problem by dynamic output feedback for linear multi-agent systems.
- We have provided a design method for computing distributed protocols such that the associated H_2 cost is smaller than a given upper bound while the controlled network reaches synchronization.

Thank you for your attention!